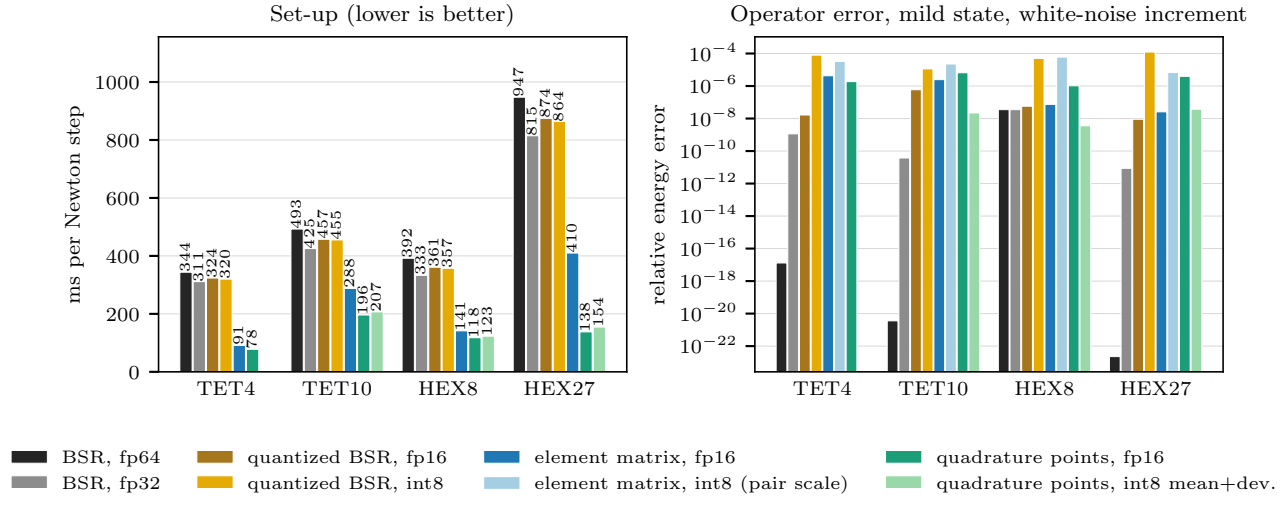

Set-up (lower is better)
ms per Newton step
0
200
400
600
800
1000
344
311
324
320
91
78
493
425
457
455
288
196
207
392
333
361
357
141
118
123
947
815
874
864
410
138
154
TET4
TET10
HEX8
HEX27
Operator error, mild state, white-noise increment
relative energy error
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
10^{-18}
10^{-20}
10^{-22}
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
quantized BSR, fp16
quantized BSR, int8
element matrix, fp16
element matrix, int8 (pair scale)
quadrature points, fp16
quadrature points, int8 mean+dev.